

AKSHATHA DUNGI

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OBJECTIVE

Software Engineer with experience building backend systems, scalable APIs, and AI-powered applications. IEEE 2026 published researcher with interests in backend infrastructure, distributed systems, and AI-integrated software.

EDUCATION

Bachelor of Technology in Computer Science and Systems Engineering

Andhra University College of Engineering, Visakhapatnam

2022 – 2026

CGPA: 9.07/10

- **Achievement:** Academic Excellence Award – Highest CGPA in CSE Department (2026)
- **Publication:** "*AI Confidence Calibration Failure Detection*" – IEEE 2026, co-author research on confidence calibration across 5 neural network architectures. Documented technical findings and presented conclusions for peer review and publication.

SKILLS

Languages	C++ (Proficient), Python (Proficient), SQL (Intermediate), Java (Beginner)
Frontend/Backend	React, HTML5, CSS3, FastAPI, Node.js, Express.js, RESTful APIs
AI/ML	TensorFlow, PyTorch, Scikit-learn, LLM Integration (Gemini API, OpenAI API), RAG, Vector Search (pgvector, FAISS)
Tools/Documentation	Git, GitHub, Docker, CI/CD

EXPERIENCE

Summer Research Intern, ISI Kolkata.

Jun 2025 – Aug 2025

Built ANN achieving 97% multi-class accuracy and 15% improvement over baseline and developed evaluation and visualization pipelines using TensorFlow, Scikit-learn, Pandas and Matplotlib.

PROJECTS

Remi – Production-Ready AI Platform

([GitHub](#))

- Architected and deployed a full-stack AI chatbot platform using React, FastAPI, PostgreSQL, JWT, and SSE.
- Built a RAG pipeline with OCR, Gemini embeddings, and pgvector similarity search for document question answering.
- Engineered secure backend APIs with authentication, validation, rate limiting, audit logging, and structured error handling.
- Automated testing and deployment with 191+ pytest tests, GitHub Actions CI/CD, Railway, and Vercel.

AI Confidence Calibration Failure Detection

(Published – IEEE 2026)

- Developed an ML evaluation framework comparing five neural architectures across 240 experiments on 160K financial records../
- Achieved 0.0032 ECE (MLP) and 0.983 accuracy across 240 experiments; discovered self-recovering calibration behavior in Residual MLP, a previously unreported finding
- Established Gradient-ECE correlation ($r=0.568$) as a novel early-warning signal for calibration failure.

LEADERSHIP

Led a five-member research team coordinating experimentation, model evaluation, and IEEE publication.